

Peer response instructions (Due Sunday by 11:59 PM). Reply to at least two classmates using 2–3 sentences per response. Provide meaningful feedback by highlighting a key strength in their response or gently explaining any concepts they could clarify.

Note: Short comments like "Great post!" or "I agree!" will not receive credit.

Reply



Jung Min Son (<https://northeastern.instructure.com/courses/271875/users/78941>)

Sep 28 9:45pm



One-Way ANOVA is useful when comparing one independent variable across three groups because it evaluates whether there is an overall difference among all group means at once. For example, if we wanted to compare average test scores among students using three different study methods, ANOVA would determine whether at least one group's mean score differs from the others. Running separate t-tests for each pair of groups may seem reasonable, but it increases the chance of making a Type I error, or incorrectly finding a statistically significant difference when one does not actually exist. This is known as the familywise error rate. The more tests that are conducted, the greater the overall risk of a false positive.

If the ANOVA result is statistically significant ($p < 0.05$), it tells us only that at least one group differs from another; it does not identify which specific groups are different. A post-hoc analysis, such as Tukey's Honestly Significant Difference (HSD) test, is necessary to compare the groups pair by pair while controlling for the increased Type I error risk created by multiple comparisons. For instance, Tukey's HSD could show if Study Method A differs from Method B, Method C, or both. Therefore, ANOVA provides the overall result, while post-hoc testing provides the specific information needed to interpret and understand it.

Reply | 



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Sep 26 4:17pm | Last reply Sep 28 9:46pm



1. When I first saw this, I was thinking why not just run t-tests on every pair. With 3 groups, that's only 3 tests (A vs B, B vs C, A vs C), which doesn't sound like a big deal. The catch is that every test comes with its own 5% chance of a Type 1 error, where we conclude there's